

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2461050.9385 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.025
Transit depth (Rp/Rs)^2	1.56 +/- 0.61 [%]
Orbital Inclination (inc)	89.8 +/- 1.42
Airmass coefficient 1 (a1)	1.1 +/- 0.022
Airmass coefficient 2 (a2)	-0.0759 +/- 0.0089
Scatter in the residuals of the lightcurve fit is	2.04 %
Best Comparison Star	None
Optimal Aperture	4.17
Optimal Annulus	7.99
Transit Duration (day)	0.0943 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.125 $\pm$ 0.025 vs 0.1406 (deviation 0.6 σ)
Duration fit vs published	0.0943 d vs 0.0975 d (deviation 0.54 σ)
Mid-time O-C	-13.1 min
Depth SNR	4.8
Residual scatter	2.04 %

Light curve

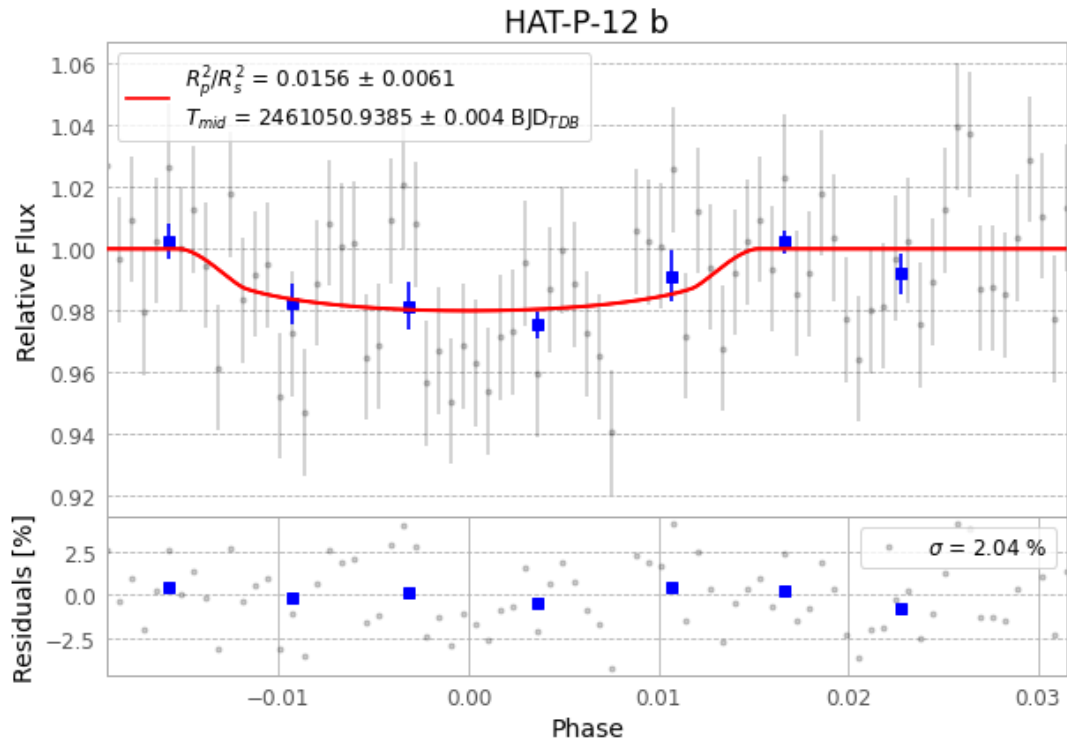


Figure 1 — FinalLightCurve_HAT-P-12 b_2026-01-10.png (SHA-256 85fe675313b27aaa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [456, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7da291f0fb0bc31b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

79 FITS frames (52 MB), HATP-12260110090016.FITS → HATP-12260110125417.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-260110.log (c076904be4ed...), FinalLightCurve_HAT-P-12 b_2026-01-10.png (85fe675313b2...), FinalLightCurve_HAT-P-12 b_2026-01-10.csv (8a3846387075...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 21:19Z.